Time-series analysis with Gaussian processes

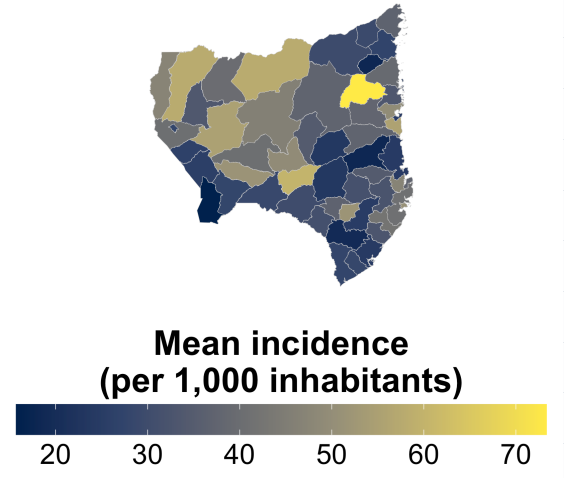
Part I: motivation, concepts and exploratory analysis

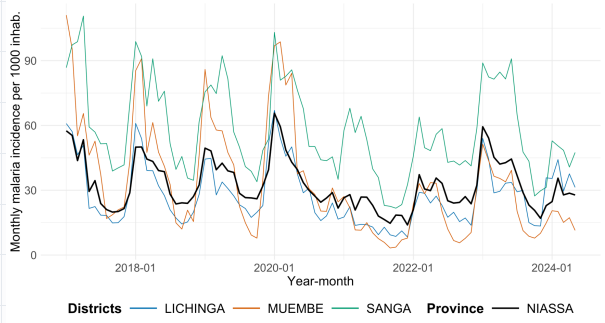
Alejandro Rozo P.

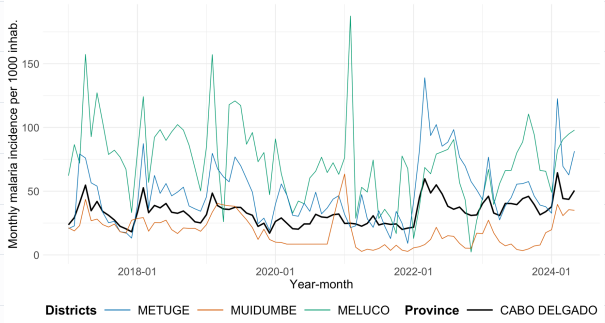


https://github.com/jarozop77/Introduction-to-time-series-analysis-with-Gaussian-Processes

Motivation: malaria surveillance in 3 provinces in Mozambique

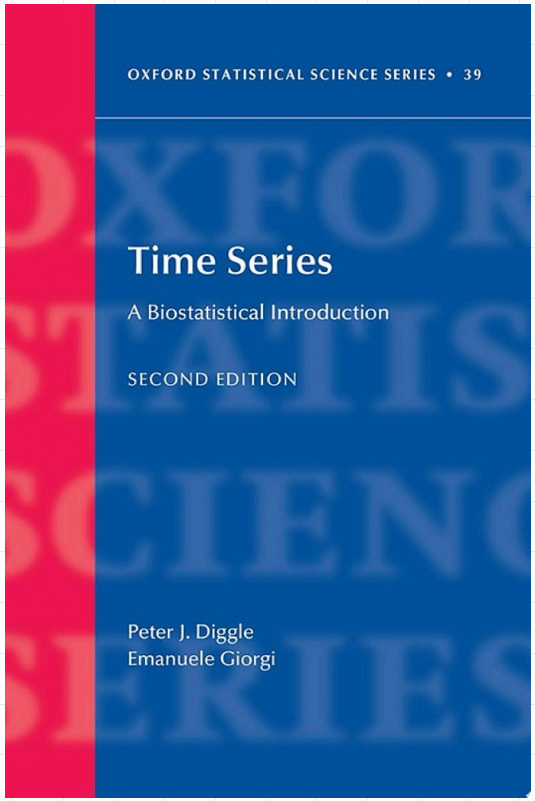






- Extend spatial models into Spatio-temporal models.
- Work with time series usually longer that the text examples.
- Interest on how to exploit the use of Gaussian processes in time series analysis

Reference for the workshop



Time Series:
A Biostatistical Introduction

Peter J. Diggle and Emanuele Giorgi
Second edition, 2024
Oxford University Press

Diggle and Giorgi (2024), 2nd edition. Oxford University Press.

Objectives of time-series analysis

- **Description:** Summarize temporal patterns and variability.
- **Forecasting and monitoring:** Predict future observations and detect changes.
- **Inference:** Estimate mean trajectories and associations with valid uncertainty.
- **Scientific understanding:** Relate temporal variation to the process generating the observations.
  - Longitudinal data
  - Repeated measures

What is a time series?

$$\{(t_i, Y(t_i)) : i = 1, \ldots, n\}, \qquad t_1 < \cdots < t_n$$

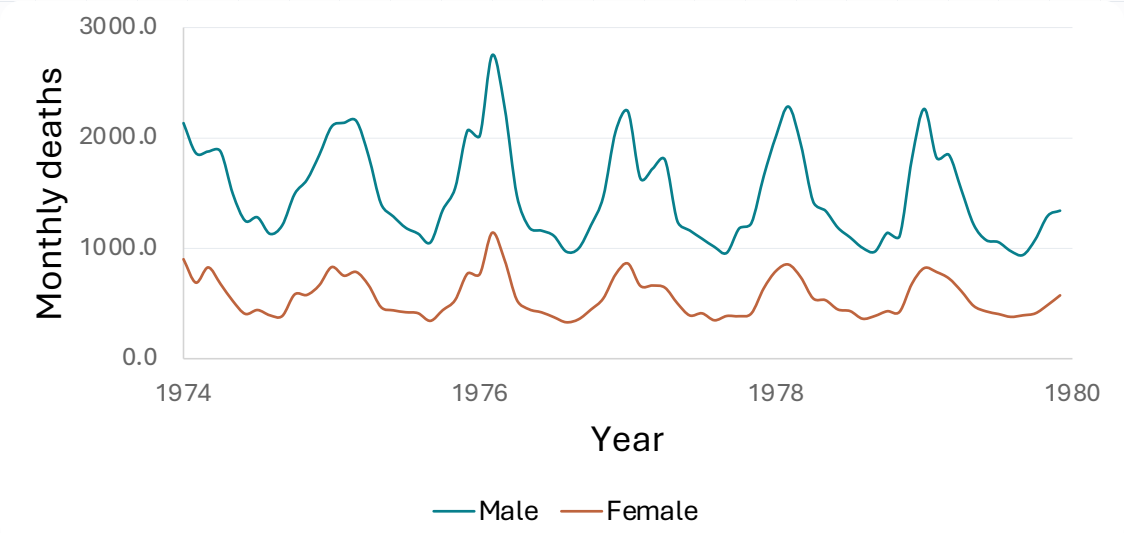
A sequence of realized values of random variables indexed by time

Three different ways finite time series can arise:

- Underlying time-scale is genuinely discrete (rare but possible)
- Time series sampled at equally spaced time points
  - Monitoring renal function in patients at risk of kidney failure (collection point every year.)
- Time series accumulated over equal intervals of time

Examples

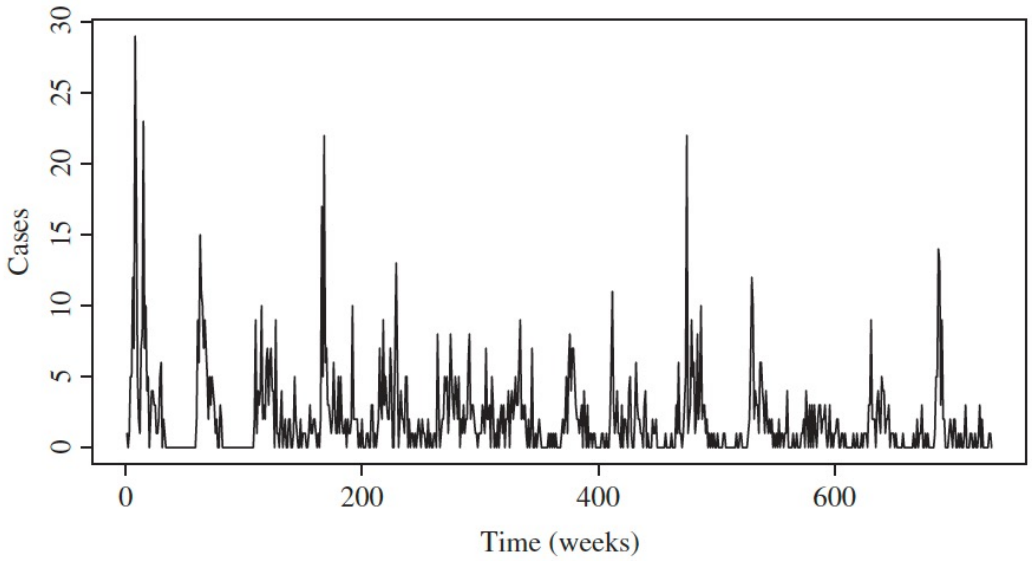
Monthly respiratory deaths in the UK



Time series accumulated over equal intervals of time

Examples

Weekly leptospirosis cases



Salvador, Brazil
1996–2010

Small counts, including zeros

Weather and long-term risk

Trend, serial dependence & stationarity

Key concepts to understand the probabilistic structure of time-series data.

Trend: the mean function

$$\mu(t) = \mathrm{E}\{Y(t)\}$$

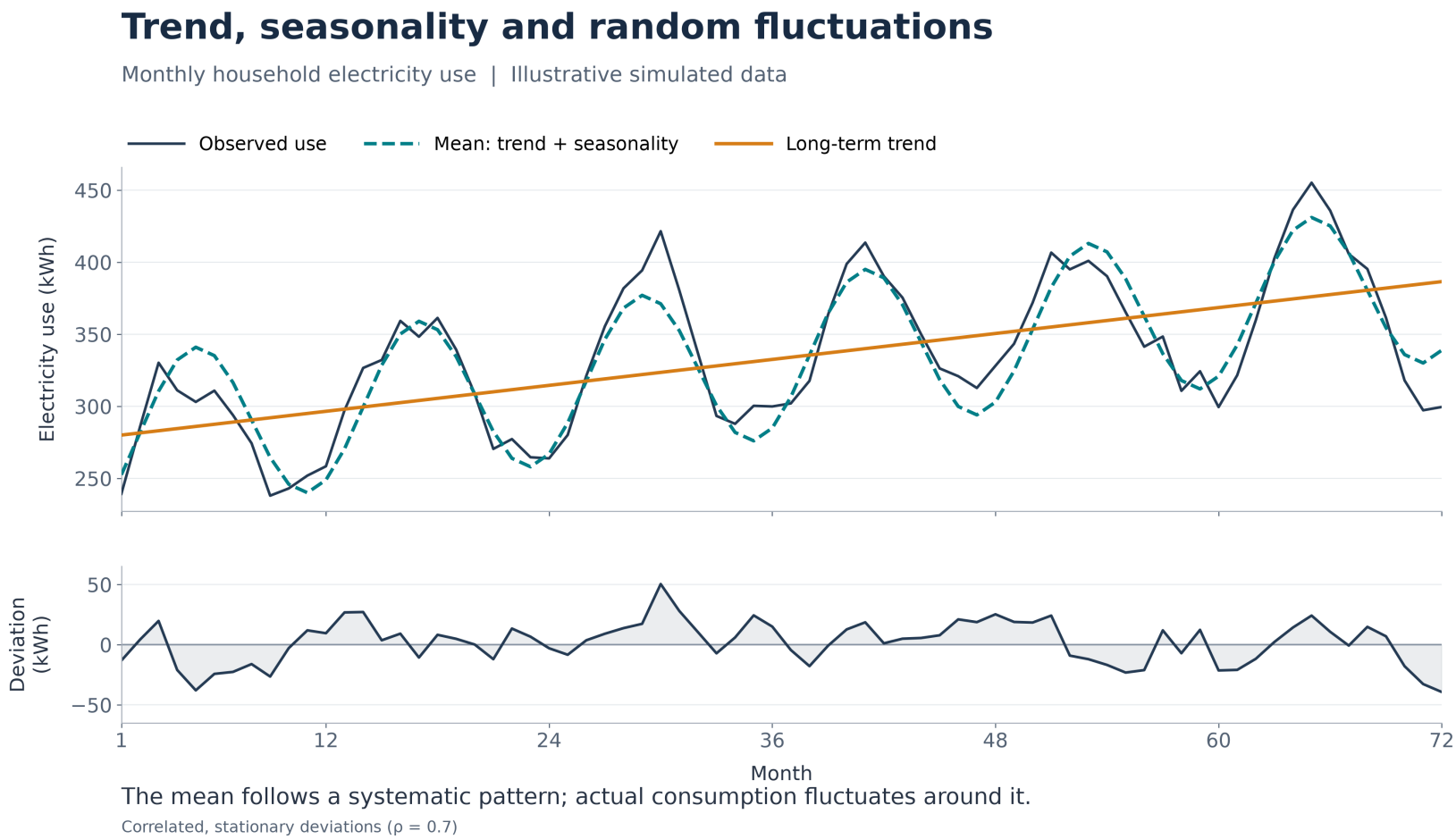
- Deterministic part.
- Long-term component.
- Includes systematic changes and seasonal mean patterns.

Serial dependence: observations at different times are dependent

$$\gamma(t, s) = \mathrm{E}[\{Y(t) - \mu(t)\}\{Y(s) - \mu(s)\}]$$

The autocovariance summarizes the strength of linear dependence.

Example: Monthly electricity consumption in a household



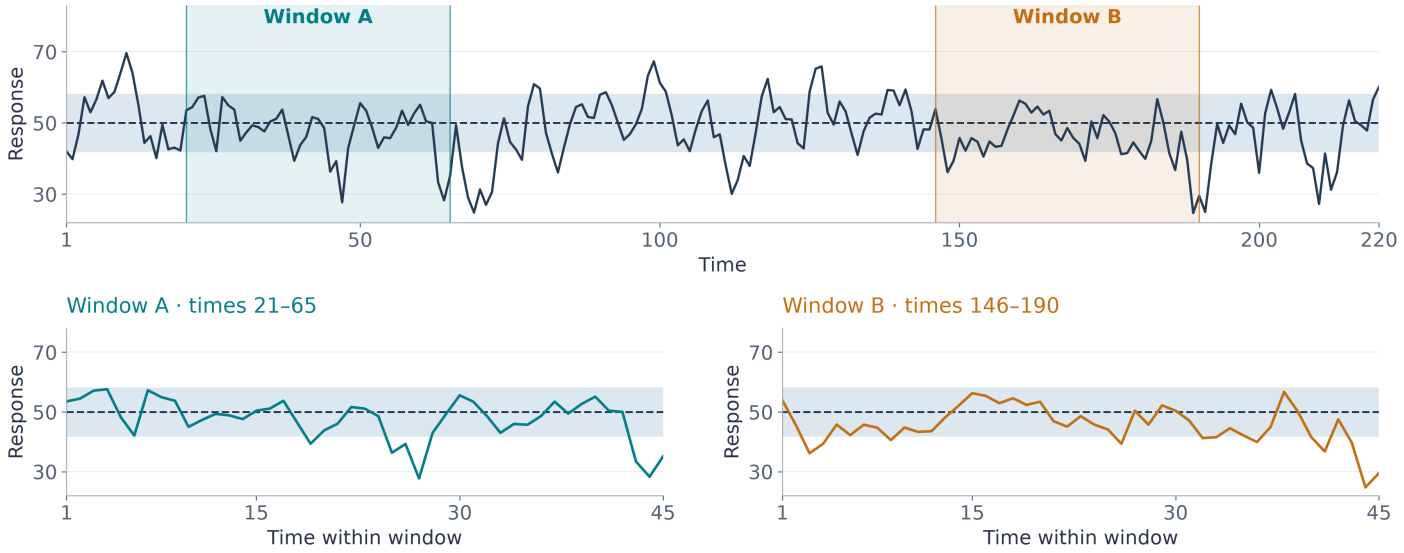
The mean follows a systematic pattern; actual consumption fluctuates around it.

Correlated, stationary deviations (ρ = 0.7)

Stationarity of the residual process

$$\mathrm{E}\{R(t)\} = 0, \quad \mathrm{Var}\{R(t)\} = \gamma(0) < \infty, \quad \mathrm{Cov}\{R(t), R(s)\} = \gamma(|t - s|)$$

- The mean and variance stay constant.
- Covariance depends on separation, not calendar time.



Same process mean and variance at every time—not identical observations or sample statistics.

Discrete time and continuous time

$$\{Y_t : t \in \mathbb{Z}\}$$

Discrete-time models

Daily malaria counts indexed by day

An AR model for monthly log pollution

$$\{Y(t) : t \in \mathbb{R}\}$$

Continuous-time models

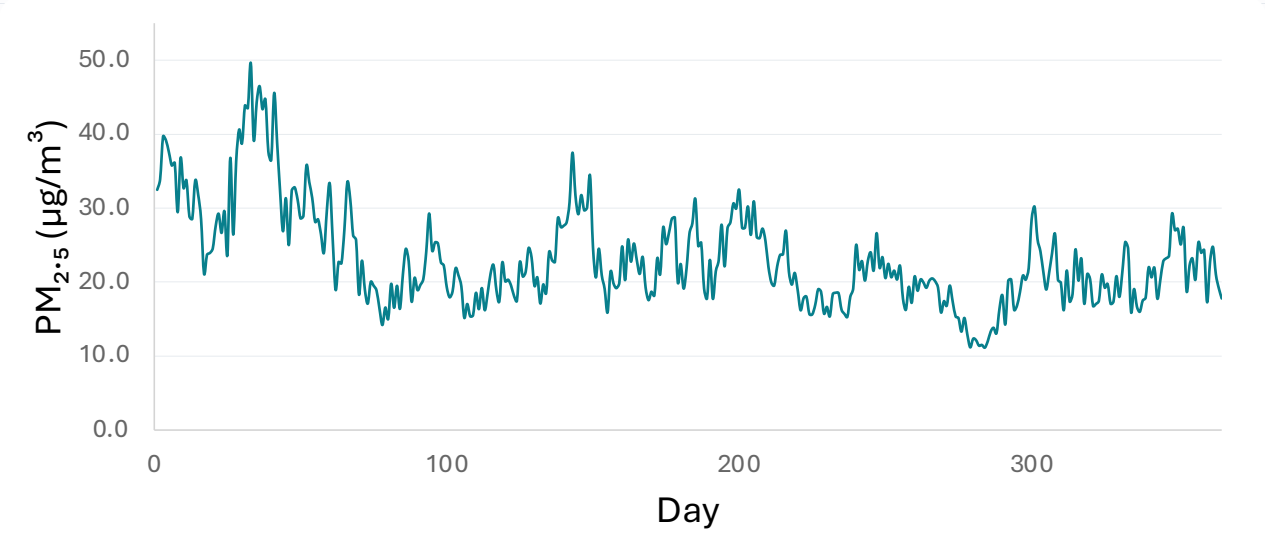
Pollution evolving continuously, measured daily

A biomarker measured at irregular visits

The time index and the outcome's measurement scale are separate choices.

Workshop example: daily air pollution

One station, 365 daily observations



Statistical objectives

Estimate systematic change.

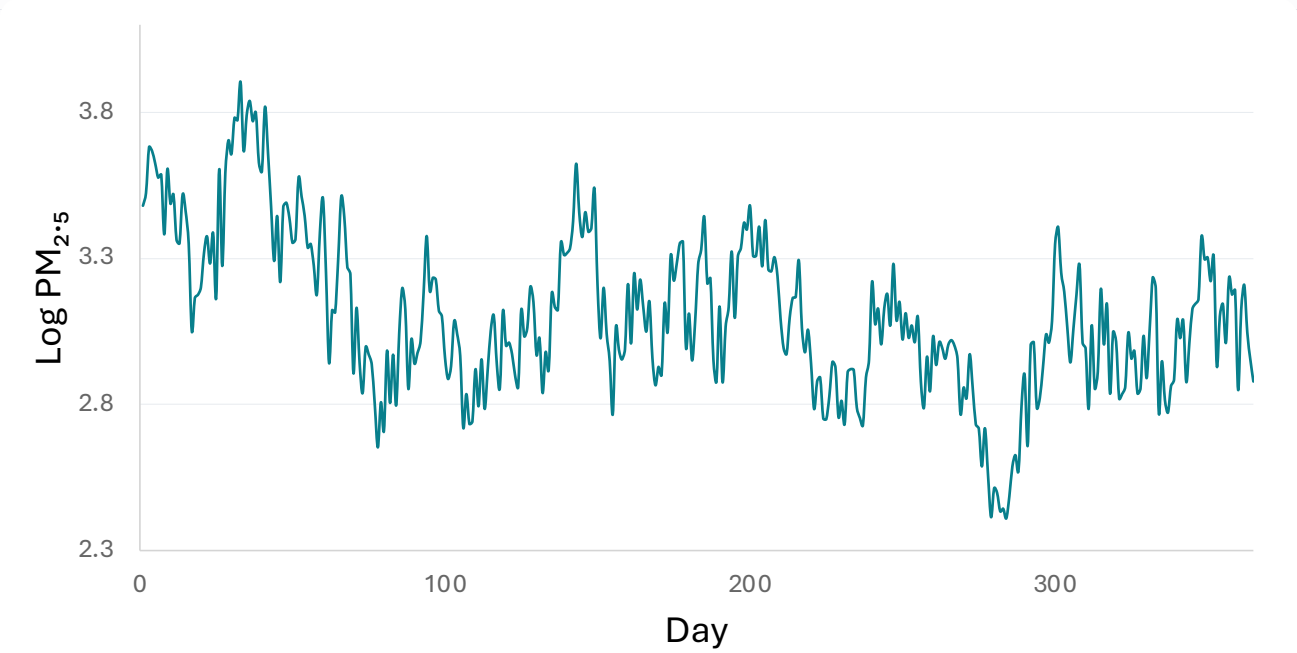
Quantify residual dependence.

Predict during monitoring gaps.

$$\log\{\mathrm{PM}_{2.5}(t)\} = \beta_0 + \beta_1 t + W(t) + \epsilon(t)$$

Exploratory data analysis.

1. Plot variable vs time



Features to inspect

Trend

Seasonality

Unusual values

Changing variability

Missing periods

A pattern in one realization is a question for the model.

2. Exploration of temporal variation decomposition

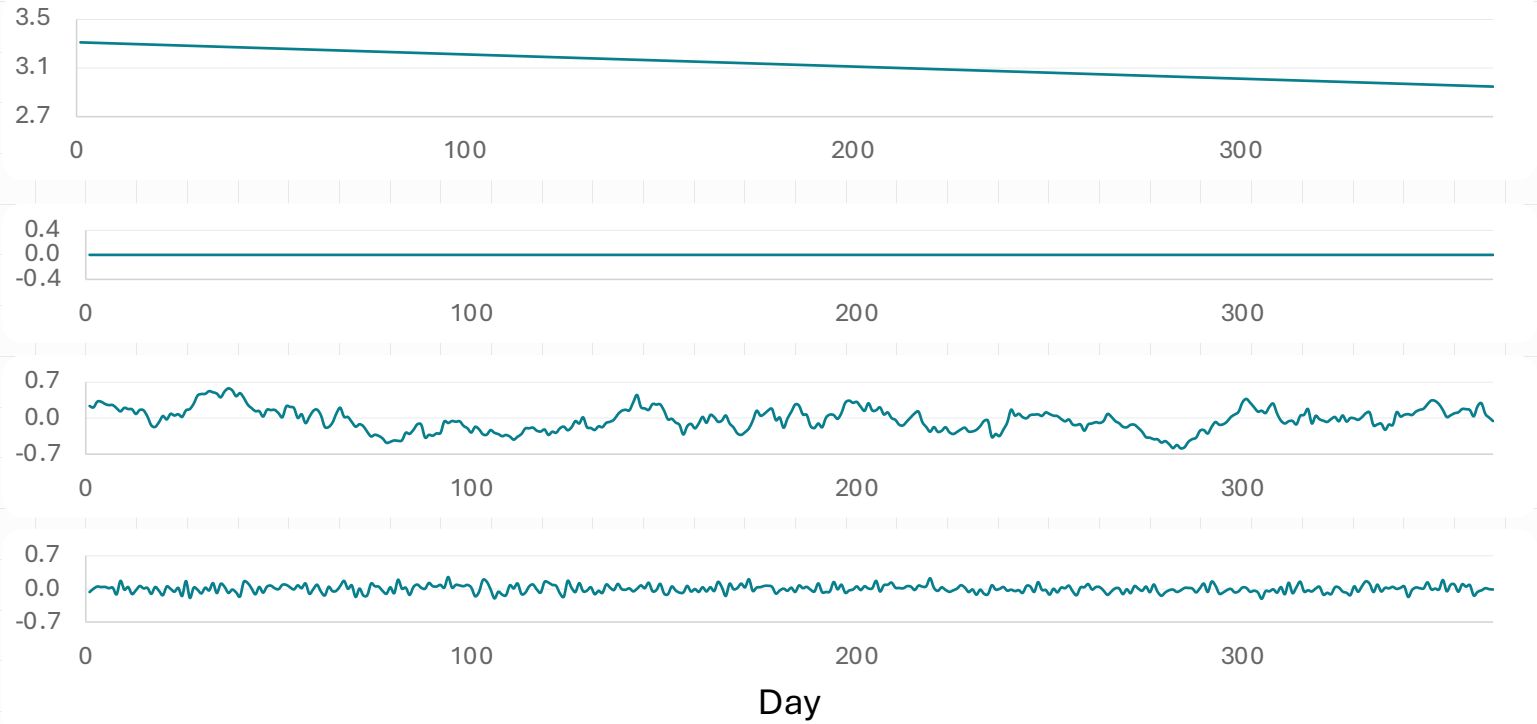
$$Y(t) = \underbrace{T(t) + S(t)}_{\text{systematic mean}} + \underbrace{W(t) + \epsilon(t)}_{\text{residual variation}}$$

Trend

Seasonality (zero)

Correlated variation

Noise



Usually reported as reminder (or random)

3. Exploration of temporal autocorrelation

A useful modelling decomposition

$$Y(t) = \underbrace{\mu(t)}_{\text{Systematic component}} + \underbrace{R(t)}_{\text{Stochastic component}}$$

$$\gamma_k = \mathrm{Cov}(R_t, R_{t-k}), \qquad \gamma_0 = \mathrm{Var}(R_t)$$

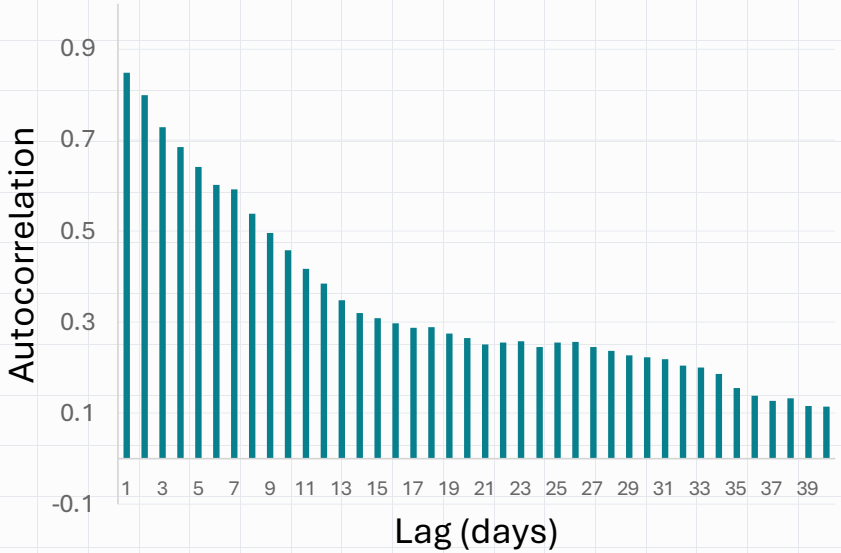
$$\rho_k = \frac{\gamma_k}{\gamma_0}, \qquad k = 0, 1, 2, \ldots$$

Autocorrelation standardizes covariance.
It has no measurement units.

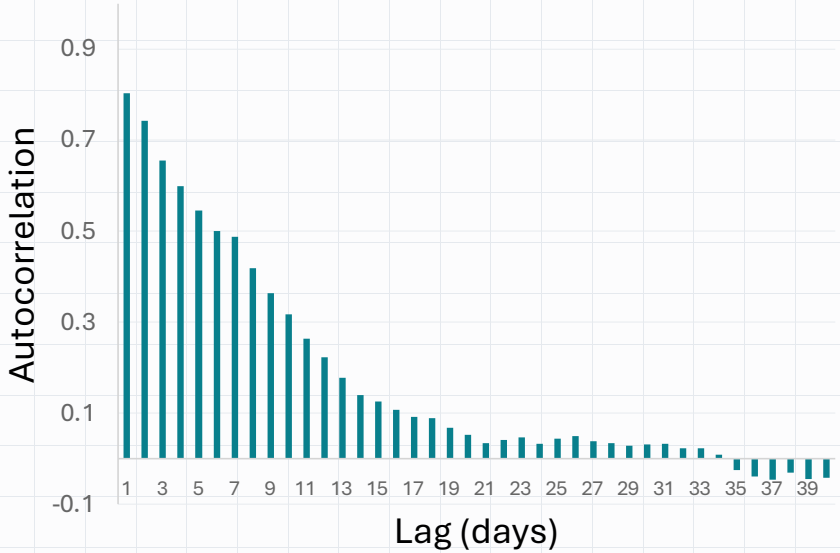
$$\widehat{\rho}_k = \frac{\sum_{t=k+1}^{n} (r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^{n} (r_t - \bar{r})^2}$$

3.1. Autocorrelation plots

Raw log concentration



After removing systematic trend



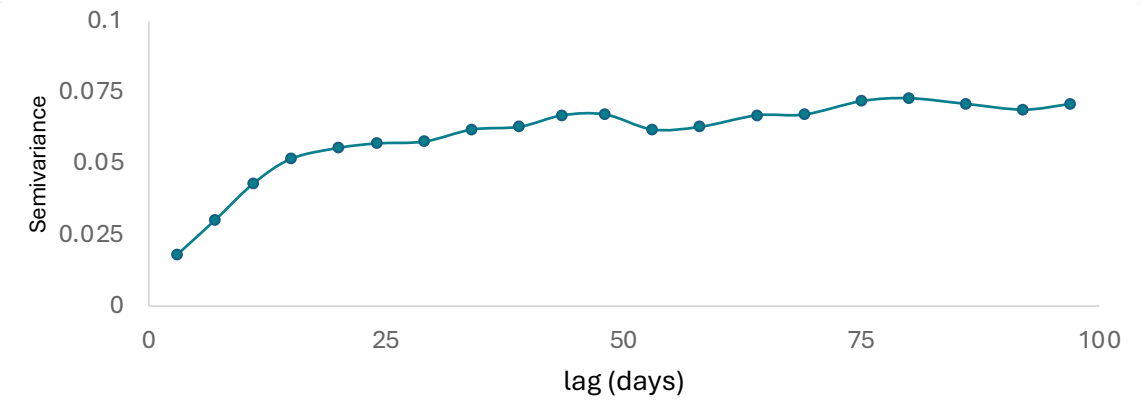
Mean specification changes the apparent residual dependence.

3.2. The (semi)variogram

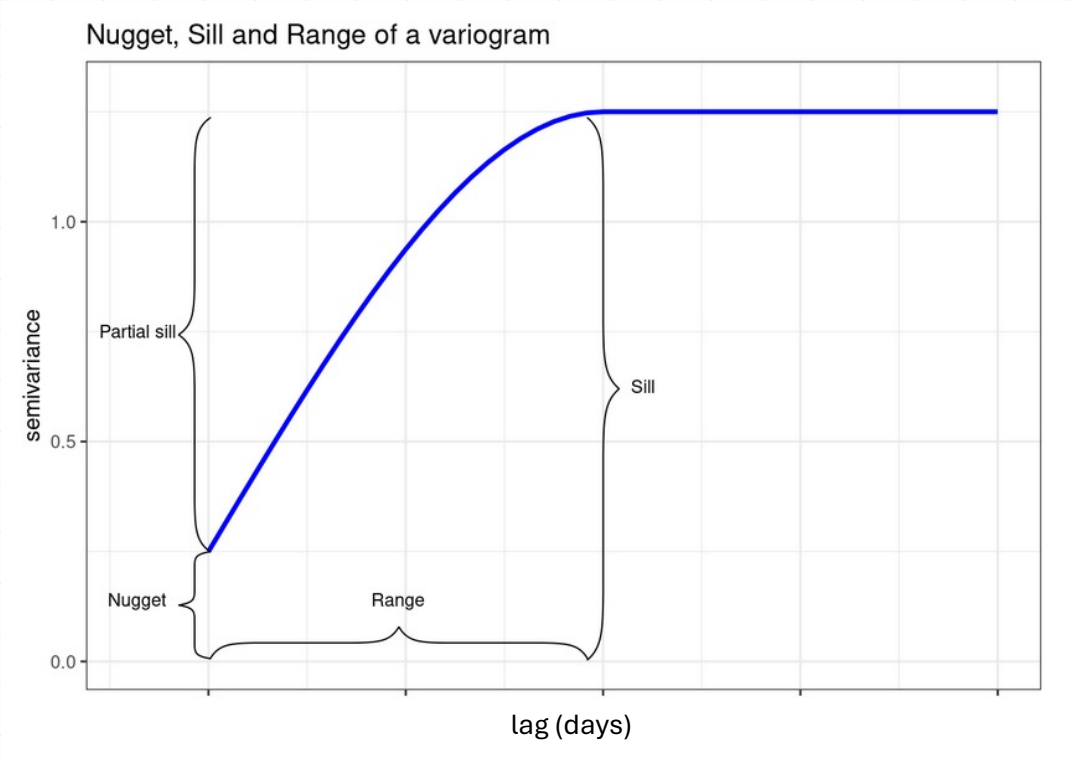
- It uses the actual time separation.
- Calculates half the average squared difference between data points separated by a specific time lag.

$$V(h) = \frac{1}{2}\mathrm{E}\left[\{R(t+h) - R(t)\}^2\right] = \gamma(0)\{1 - \rho(h)\} \qquad h_{ij} = |t_i - t_j|$$

- Nearby observations in time tend to be similar → variogram values will be low
- As lags increases, their differences grow since dependence weakens → variogram values tend to increase to the point of reaching a 'plateau'



Parameters involved in the variogram: Nugget, sill and range



Nugget

The variance at time lags close to 0. Reflecting residual variability or temporal variability at other scales

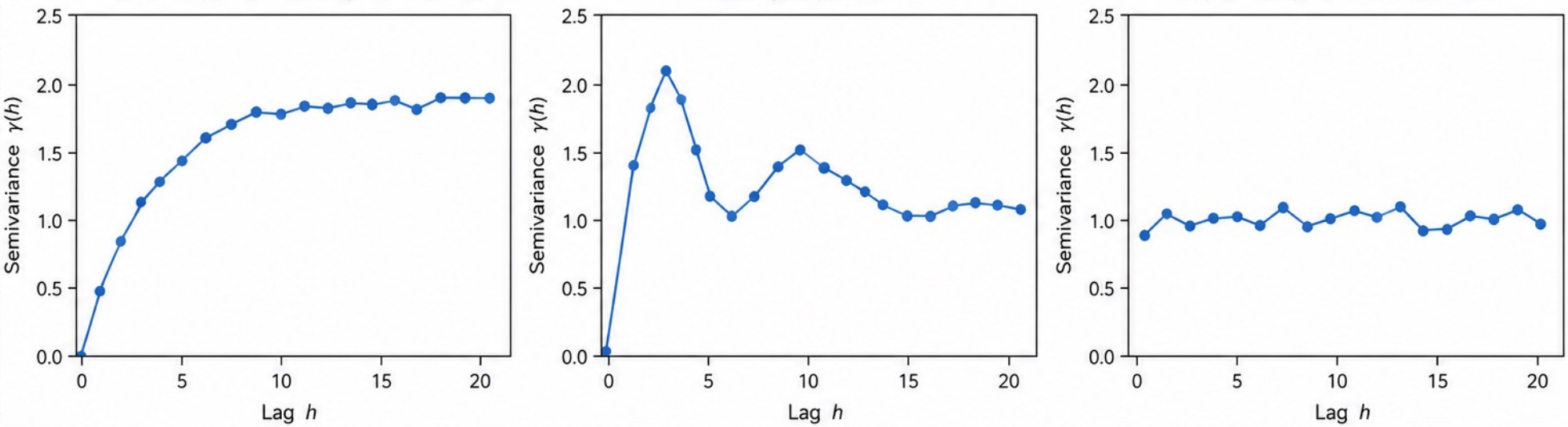
Sill

The value at which the variogram levels off, corresponding to the marginal variance for a stationary process.

Range

Time lag to which temporal dependence is negligible.

Examples



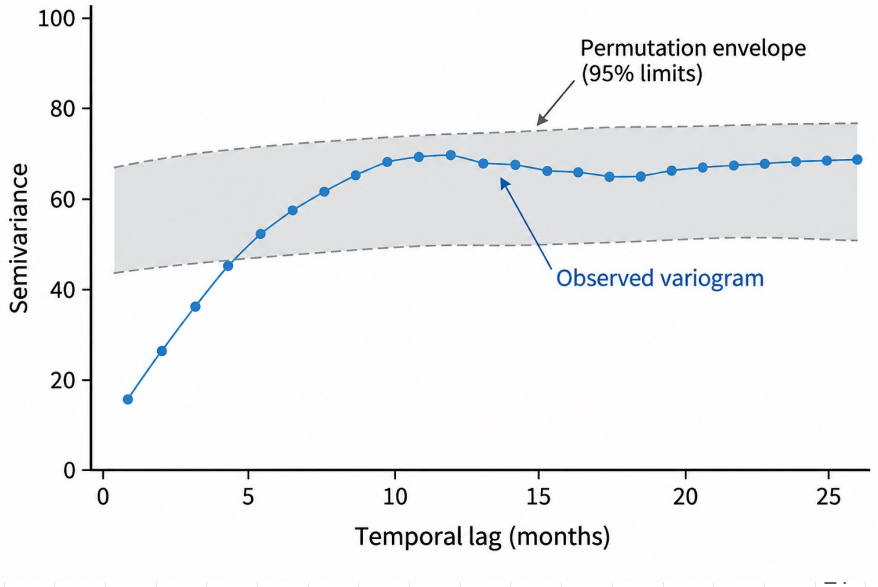
Permutation test using the variogram

$$H_0: No\ temporal\ dependence \qquad vs \qquad H_1: Temporal\ dependence\ exists$$

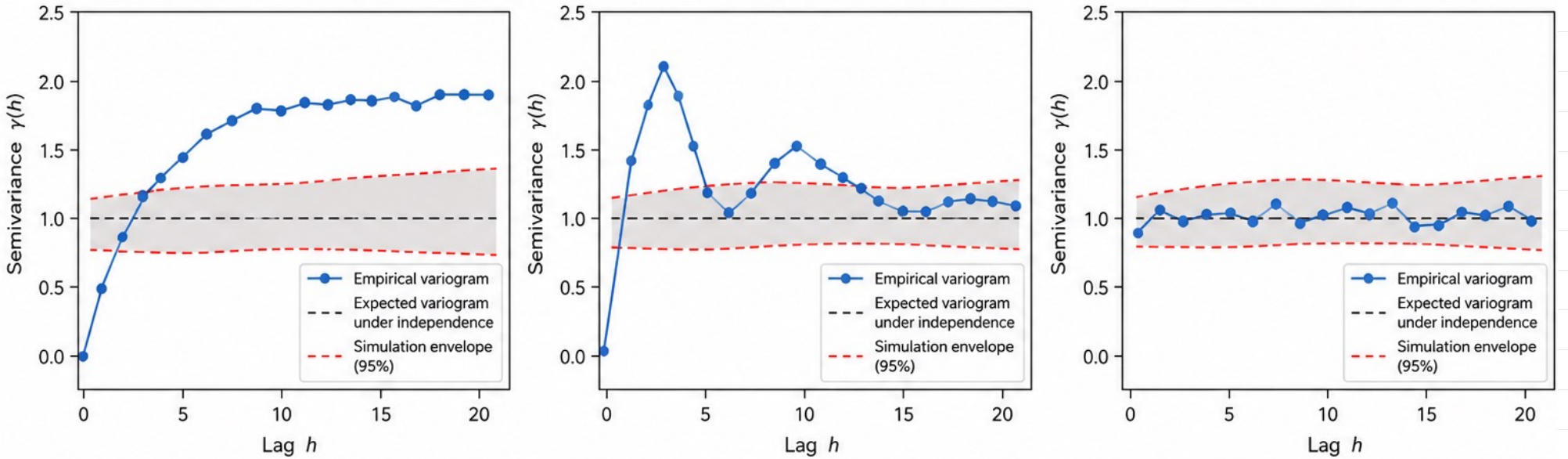
- Under null hypothesis, observations are independent and identically distributed, so their order is exchangeable.

Permutation reference envelope:

- Calculate the empirical variogram from the original time series
- Randomly permute the temporal order of the observations.
- Calculate a new variogram for each permutation.
- Repeat this process, for example, 999 times.
- Compare the observed variogram with the distribution of permuted variograms



Examples



- What is the sill, range and nugget values?

The practical range describes how long meaningful temporal dependence persists.

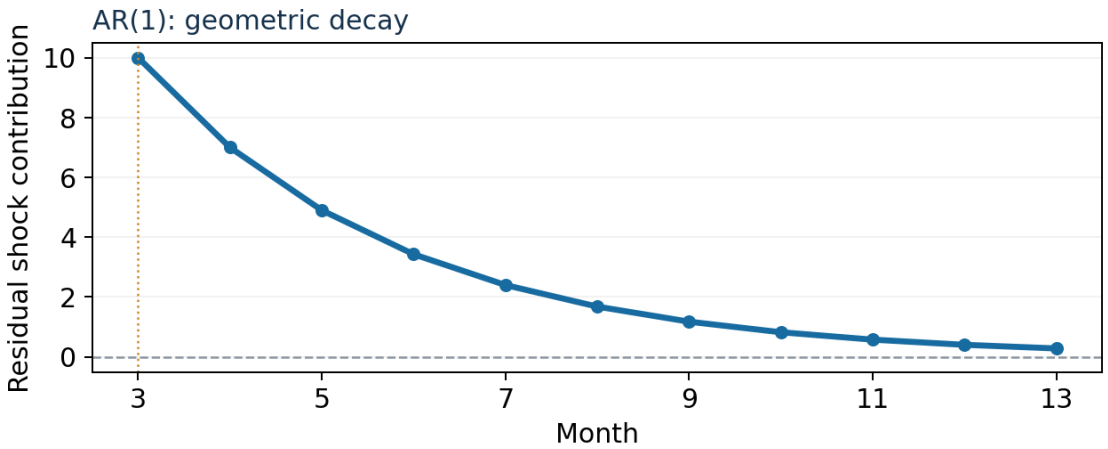
Familiar temporal models: AutoRegressive (1)

$$R_t = \alpha R_{t-1} + Z_t, \qquad |\alpha| < 1$$

$$E(Z_t) = 0$$

$$Var(Z_t) = \sigma_z^2 < \infty$$

$$Cov(Z_t, Z_s) = 0, \qquad t \neq s$$



INTUITION

A positive residual tends to persist, but its influence fades.

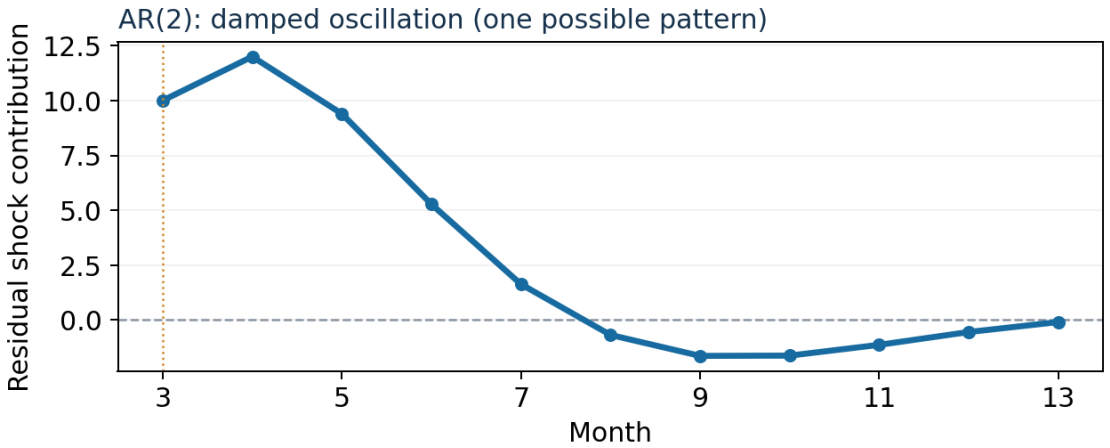
ASSUMPTION / LIMITATION

Stationary when |ϕ| < 1; short-memory geometric dependence.

Example: ϕ = 0.7; a +10 shock leaves 7, 4.9, 3.43, …

Familiar temporal models: AutoRegressive (2)

$$R_t = \alpha_1 R_{t-1} + \alpha_2 R_{t-2} + Z_t$$



INTUITION

A positive residual tends to persist, but its influence fades.

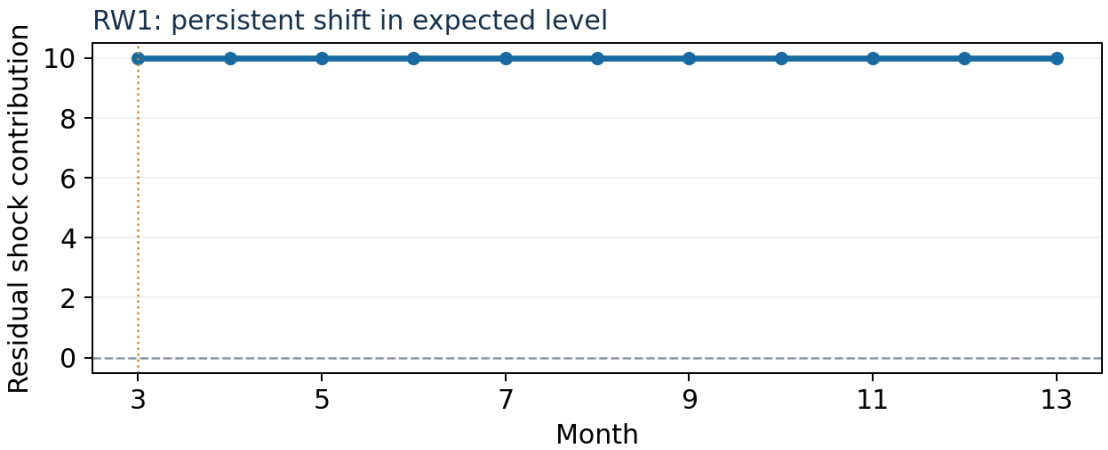
ASSUMPTION / LIMITATION

Can capture damped oscillations; requires AR(2) stationarity conditions.

Example: $\phi_1 = 1.2$, $\phi_2 = -0.5$; a +10 shock may overshoot.

Familiar temporal models: Random Walk (1)

$$R_t = R_{t-1} + Z_t, \quad R_0 = 0, \quad \mathrm{Var}(R_t) = t\sigma_Z^2$$



INTUITION

Unexpected changes accumulate instead of reverting to zero.

ASSUMPTION / LIMITATION

Nonstationary; variance increases with time; shocks persist.

Example: a +10 shock shifts the expected residual level by +10.

The connection to Gaussian processes

$$(W(t_1), \ldots, W(t_n))^\top \sim N_n(\mathbf{0}, \Sigma)$$

- Gaussian AR models and Gaussian random walks are GPs.
- A Gaussian process doesn't imply stationarity.

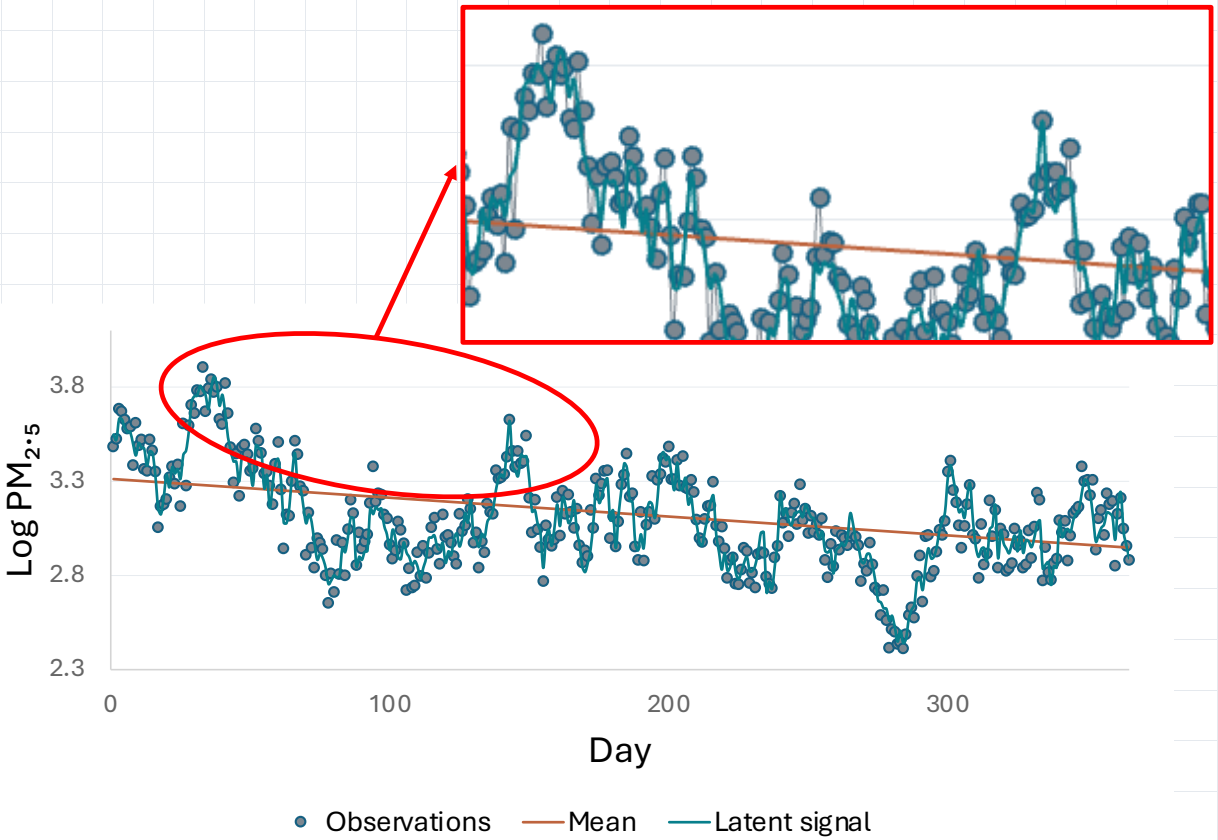
Time-series analysis with Gaussian processes

Part II: Modeling temporal structure with Gaussian Processes

Alejandro Rozo P.

A temporal model for log pollution

$$Y(t_i) = x(t_i)^\top \beta + W(t_i) + \epsilon_i$$



$Y(t_i)$ is log $PM_{2.5}$ concentration

Systematic mean

Time and measured covariates

Correlated variation W(t)

Departures that persist over time

Independent noise ε_i

Measurement error or unresolved variation

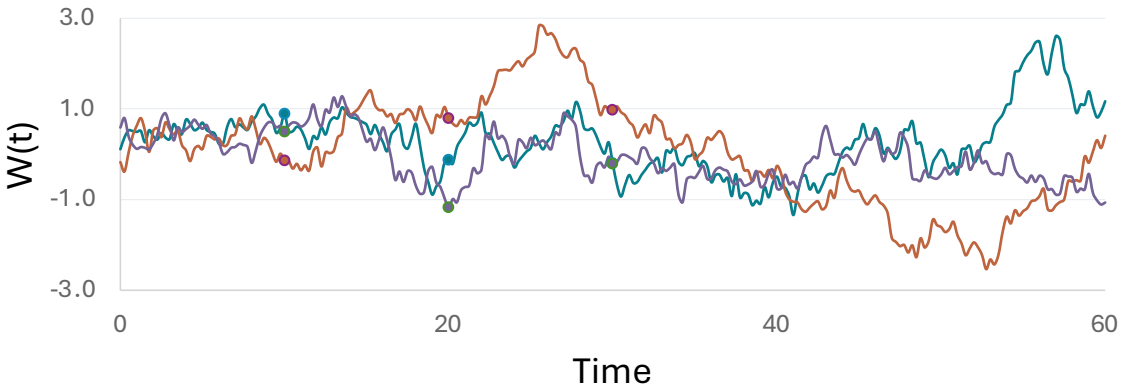
The Gaussian-process definition

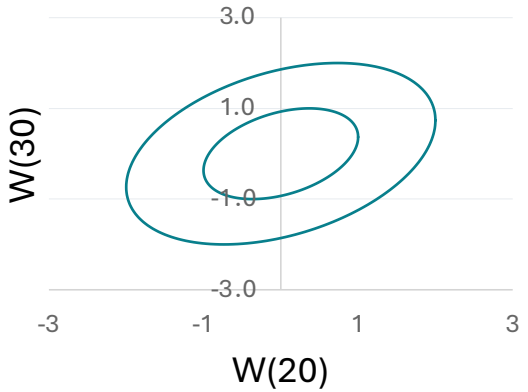
$$\begin{pmatrix} W_1 \\ \vdots \\ W_T \end{pmatrix} \sim N(\mu(t), C(t,t'))$$

$\mu(t)$: It describes the expected value at each time point

$C(t,t')$: It describes the dependence between values at different time points

Joint normality for every finite selection of times





Gaussian processes in the context of time series

$$\begin{pmatrix} W_1 \\ \vdots \\ W_T \end{pmatrix} \sim N(0, \sigma^2 C(t, t'))$$

$$Y(t_i) = x(t_i)^\top \beta + W(t_i) + \epsilon_i$$

$$\epsilon_i \overset{\text{iid}}{\sim} N(0, \tau^2), \qquad \epsilon_i \perp W$$

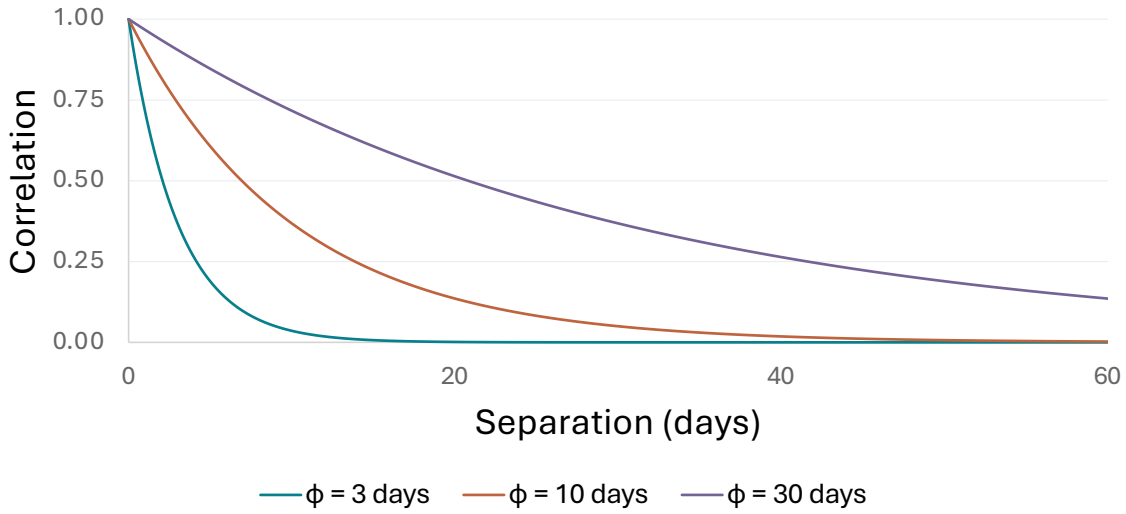
Variance	Interpretation
σ^2	Correlated process
τ^2	Independent noise
$\sigma^2 + \tau^2$	Marginal residual variance

Exponential covariance.

$$Y(t_i) = x(t_i)^\top \beta + W(t_i) + \epsilon_i$$

$$\begin{pmatrix} W_1 \\ \vdots \\ W_T \end{pmatrix} \sim N(0, C(t, t'))$$

$$C_W(h) = \sigma^2 \exp(-h/\phi), \qquad h \geq 0$$



ϕ determines over what period observations remain substantially correlated

At h = ϕ, correlation ≈ 0.37

At h = 3ϕ, correlation ≈ 0.05

It can be shown that a stationary Gaussian process with exponential covariance, observed at equally spaced times, has the same distribution as a stationary Gaussian AR(1) process

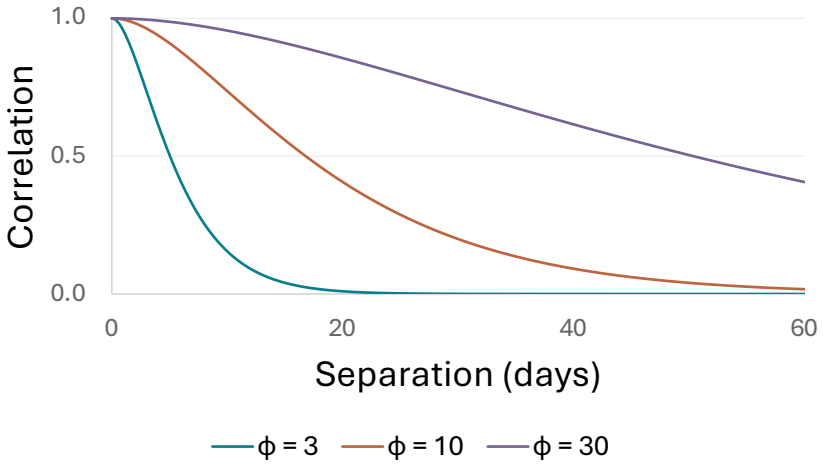
The Matérn family.

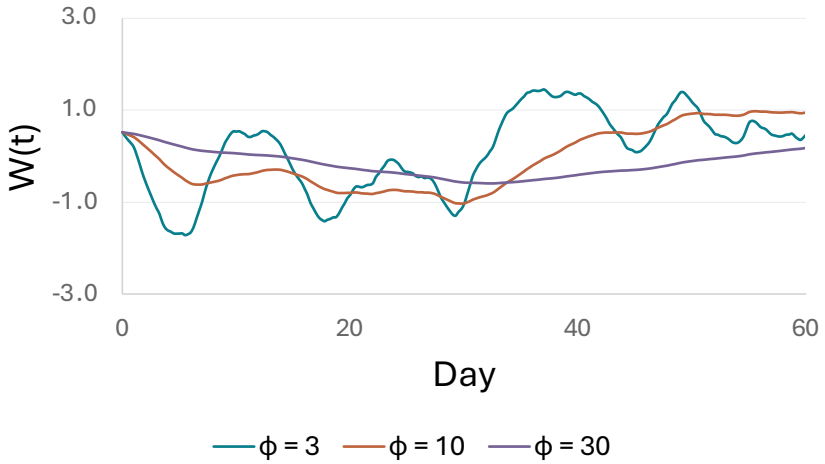
$$C_W(h) = \frac{1}{2^{\kappa-1}\Gamma(\kappa)} \left(\frac{h}{\phi}\right)^{\kappa} K_\kappa \left(\frac{h}{\phi}\right)$$

Smoothing parameter

Range parameter

Fixed smoothness κ = 1.5, varying scale ϕ

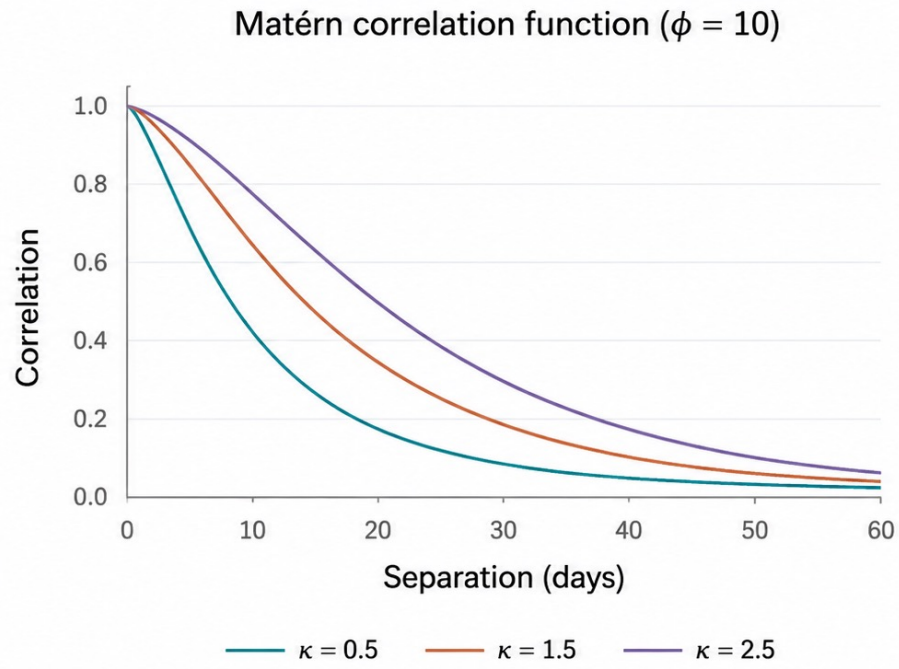


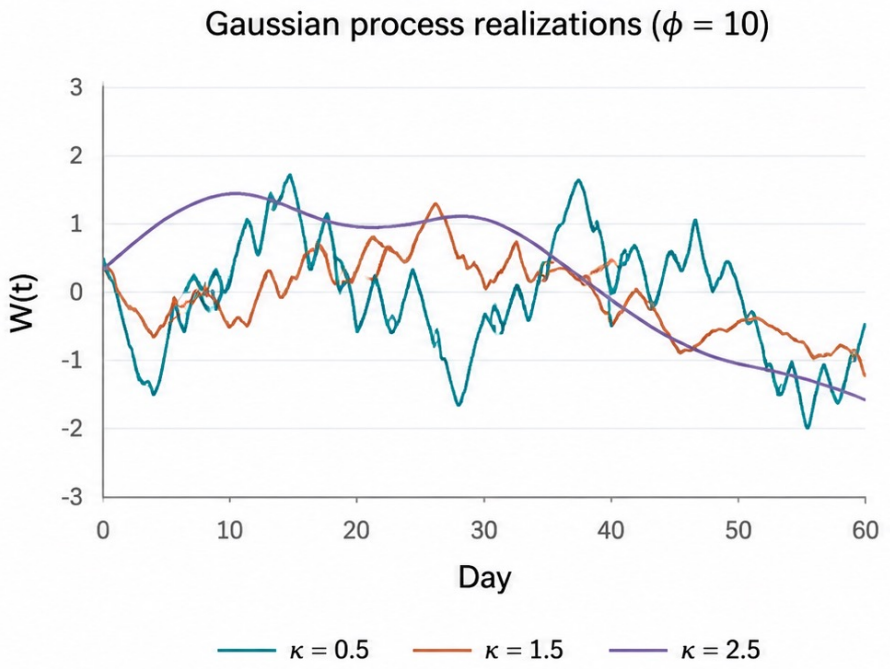


Small ϕ the process fluctuates relatively rapidly. Large ϕ changes occur over longer periods, and deviations from the mean tend to persist.

The Matérn family: process smoothness

- Smoothness is primarily about what happens between observations that are very close in time.





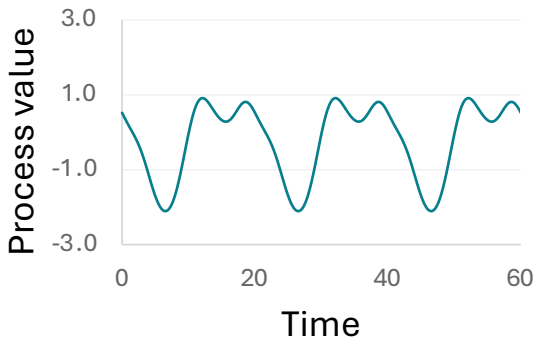
Comments about the smoothing parameter κ

- $\kappa > 0$
- Common values to choose from: 0.5, 1.5 and 2.5
- Mathematically convenient and more intuitive interpretation
- Identifiability issue among (σ^2, ϕ, κ). To circumvent this, in practice κ is fixed.

Variance	Interpretation
σ^2	Marginal variance of the process
ϕ	Practical range
κ	Smoothness of the process

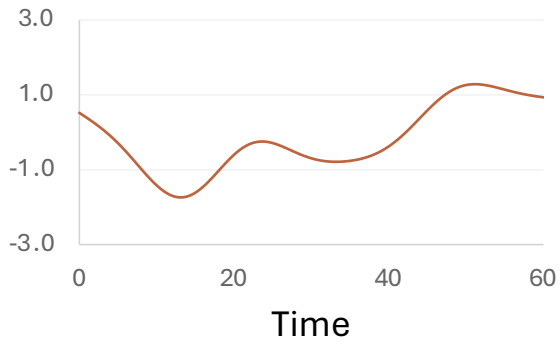
Other Gaussian-process constructions

Periodic covariance



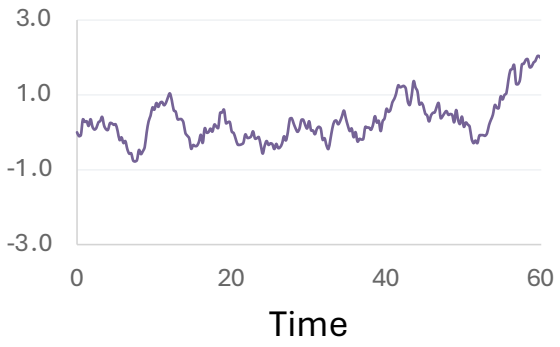
Random variation that repeats at a fixed period

Squared exponential



Very smooth latent trajectories

Brownian motion



Accumulating uncertainty and nonstationarity

These process assumptions support likelihood-based or Bayesian inference.

Maximum likelihood estimates the model jointly

$$\mathbf{Y} \sim N_n(X\beta, \Sigma), \quad \Sigma = \sigma^2 R(\phi, \kappa) + \tau^2 I$$

$$\ell(\theta) = -\frac{1}{2}\left[n\log(2\pi) + \log|\Sigma| + (\mathbf{y} - X\beta)^\top \Sigma^{-1}(\mathbf{y} - X\beta)\right]$$

$$\theta = (\beta, \sigma^2, \tau^2, \phi, \kappa)$$

- Bayesian estimation is also possible.

Maximum likelihood estimates the model jointly

$$\mathbf{Y} \sim N_n(X\beta, \Sigma), \quad \Sigma = \sigma^2 R(\phi, \kappa) + \tau^2 I$$

$$\ell(\theta) = -\frac{1}{2}\left[n\log(2\pi) + \log|\Sigma| + (\mathbf{y} - X\beta)^\top \Sigma^{-1}(\mathbf{y} - X\beta)\right]$$

$$\theta = (\beta, \sigma^2, \tau^2, \phi, \kappa)$$

- Bayesian estimation is also possible.

Conditional Gaussian prediction

$$c_* = \mathrm{Cov}(\mathbf{Y}, W(t_*))$$

$$\mathrm{E}\{S(t_*) \mid \mathbf{y}\} = x(t_*)^\top \beta + c_*^\top \Sigma^{-1}(\mathbf{y} - X\beta)$$

$$\mathrm{Var}\{S(t_*) \mid \mathbf{y}\} = \sigma^2 - c_*^\top \Sigma^{-1} c_*$$

$$\mathrm{Var}\{Y_{\mathrm{new}}(t_*) \mid \mathbf{y}\} = \mathrm{Var}\{S(t_*) \mid \mathbf{y}\} + \tau^2$$

Known parameters: exact conditional Gaussian results.
Estimated parameters: plug-in prediction intervals.

Time-series analysis with Gaussian processes

Part III: Implementations in R

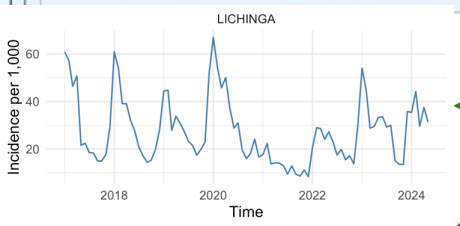
Alejandro Rozo P.

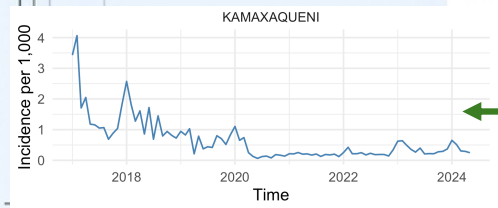
Time-series analysis with Gaussian processes

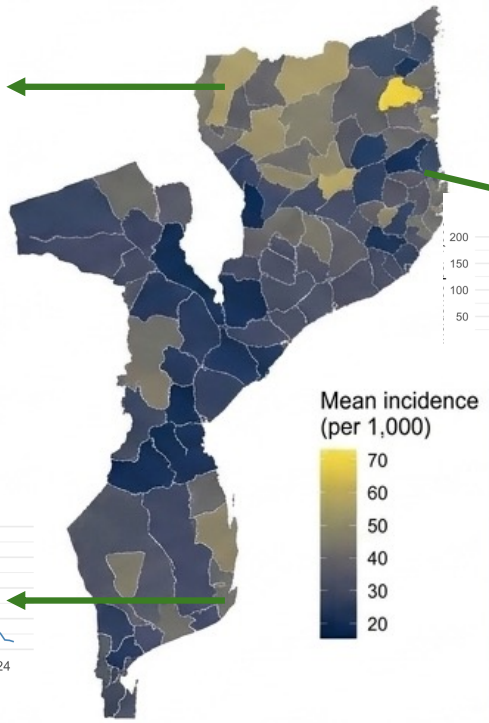
Part IV: Gaussian processes in display: Hierarchical Spatio-Temporal Modeling: Area-Specific Gaussian Processes with Spatially Shared Hyperparameter for Malaria Surveillance

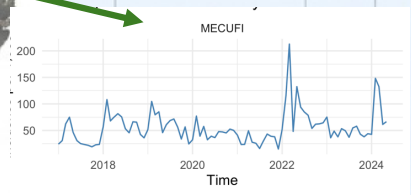
Alejandro Rozo P.

The Public Health Context









Malaria transmission exhibits spatial and temporal heterogeneity.

The Burden

Entire population at risk, 1/3 of the population infected.

Region

161 districts.

The Timeline

83 months, June 2017 to May 2024.

Covariates

Daily climatic data (temperature, precipitation, relative humidity), vegetation index, intervention data.

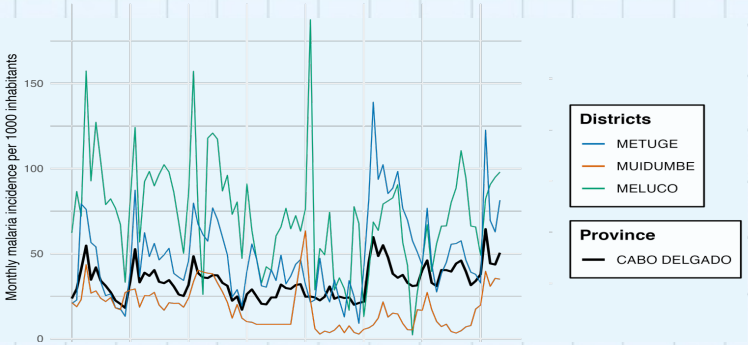
The Data Mismatch & Modeling Challenge

The Mismatch



Disease cases are reported monthly, but the environmental drivers (temperature, rainfall, …) act at higher resolutions

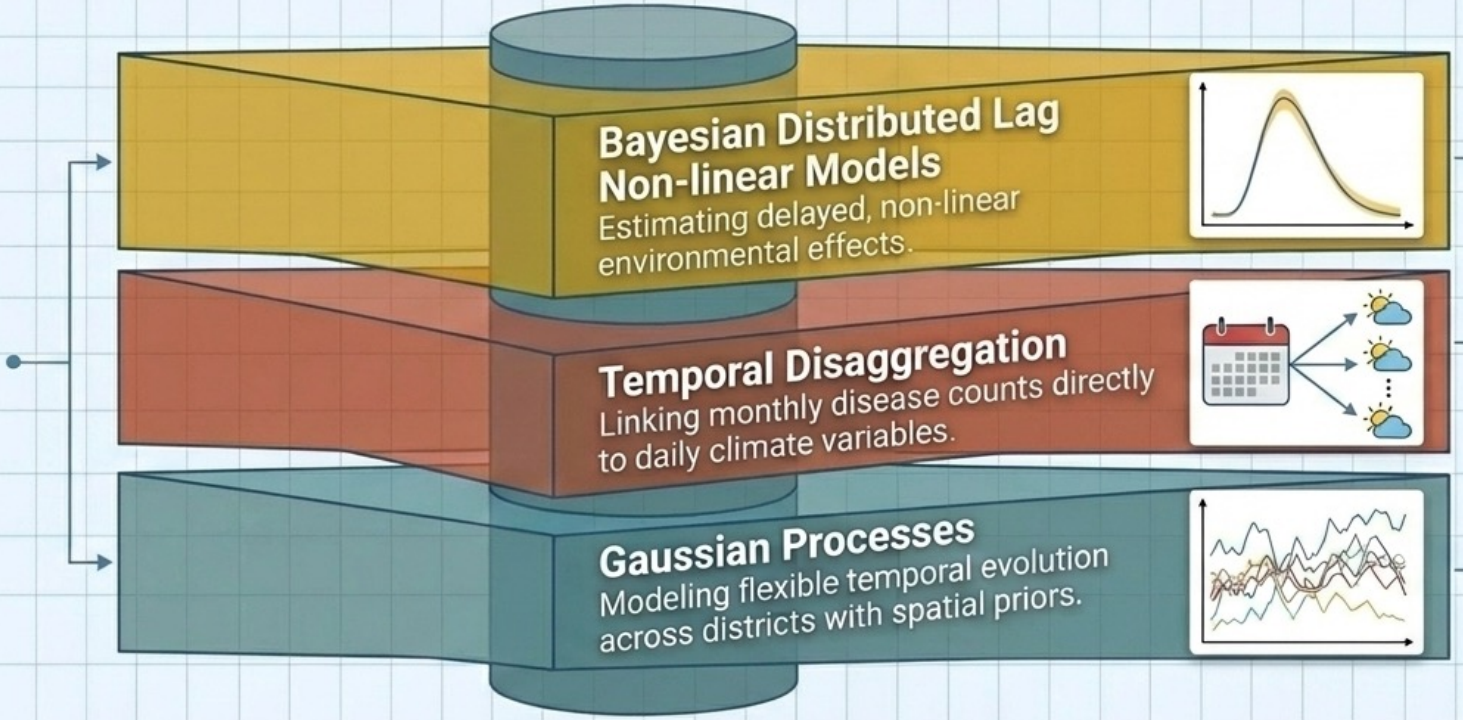
The Complexity



Allow region-specific trajectories while coherently borrowing strength across space and time.

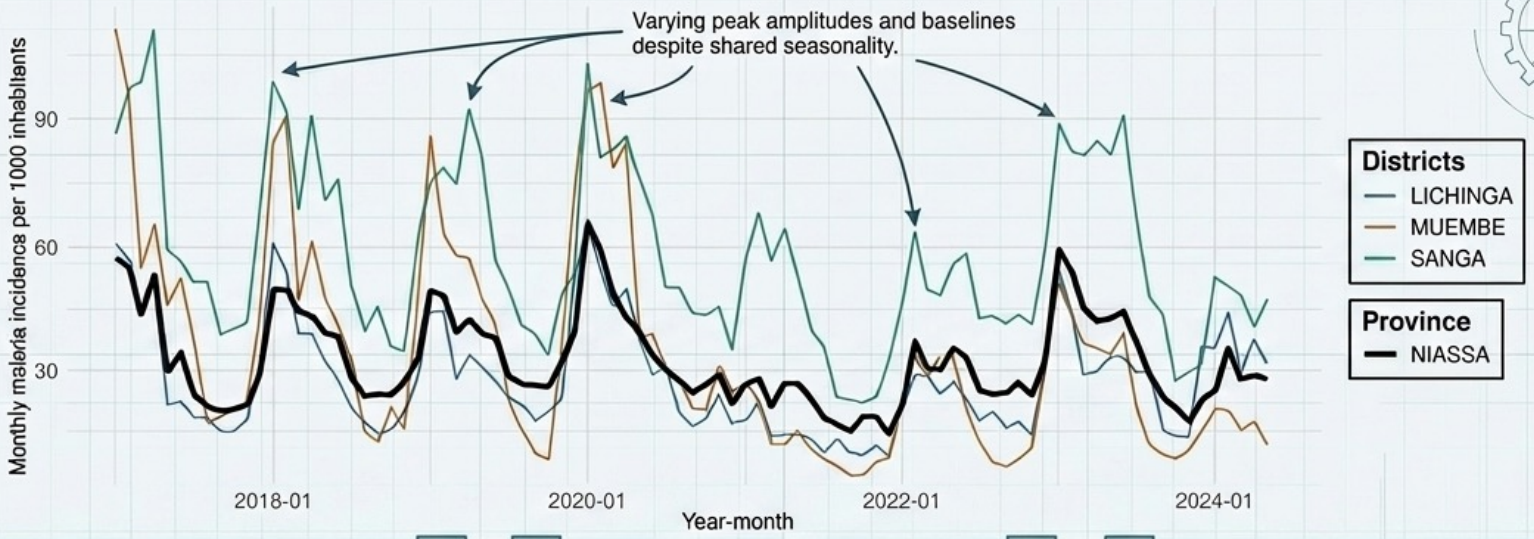
What characteristics must an appropriate statistical model have to bridge this gap?

The Integrated Bayesian Architecture



Our framework weaves each methodological component into a single hierarchical model.

Problem 1: The Assumption of Global Temporal Trends



The Limitation

Standard spatio-temporal models often impose a single, common temporal structure across all regions.

The Consequence

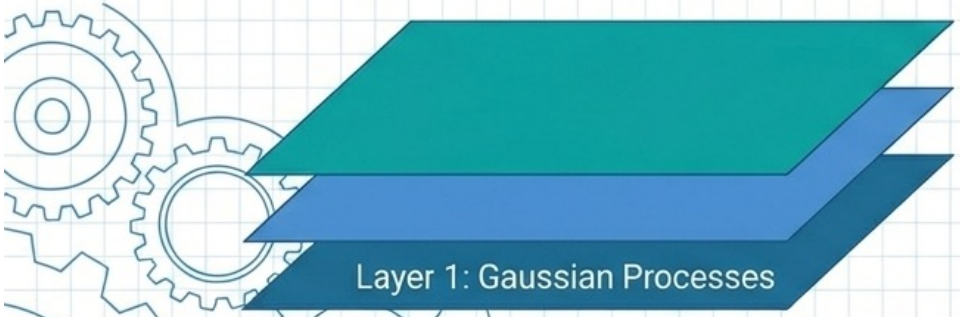
This fails to capture hyper-local outbreaks and assumes districts evolve in parallel over time.

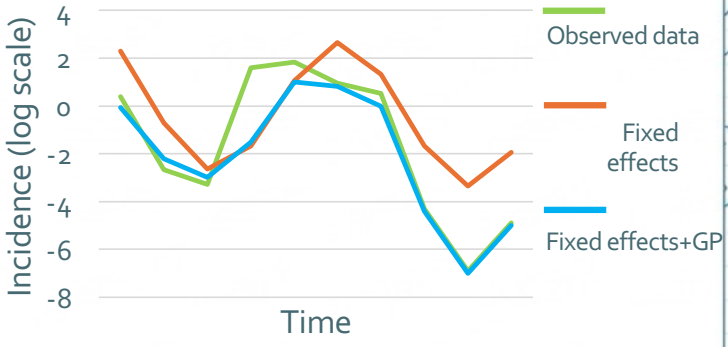
The Question

How do we model flexible temporal evolution across districts without completely losing statistical power?

Solution 1: Spatially correlated hyperparameters

- **The Mechanism**: We model each district's time series using a Gaussian Process (GP).
- **The Innovation**: We introduce spatial structure not on the outcomes, but on the GP hyperparameters (temporal variance and range).
- **The Result:** Neighboring regions share information about the characteristics of their temporal dynamics (smoothness, variability) while preserving their own specific, independent trajectories.



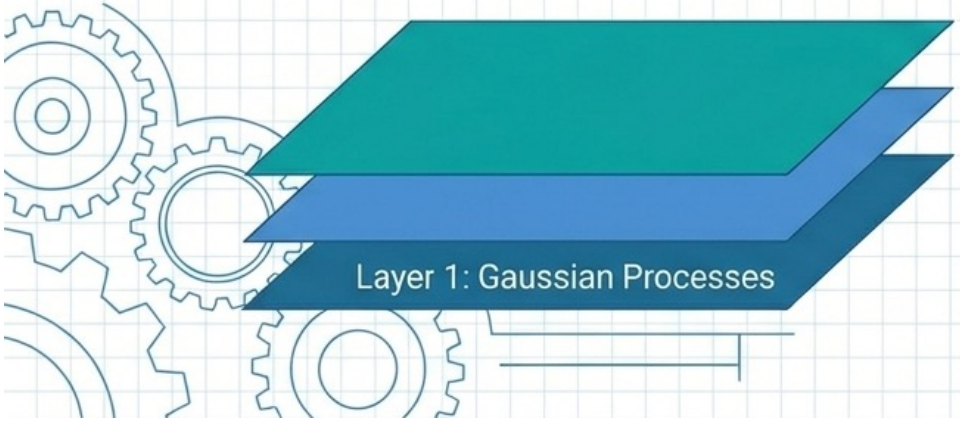


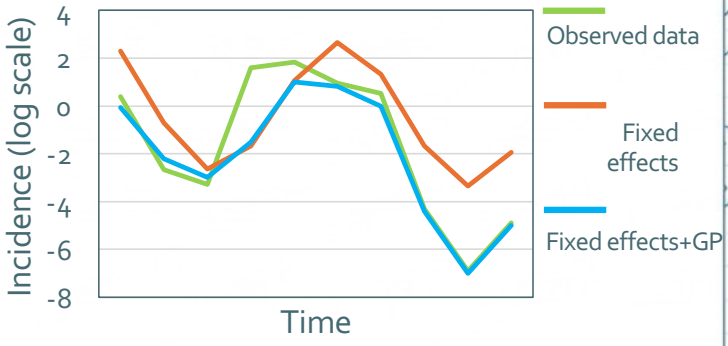


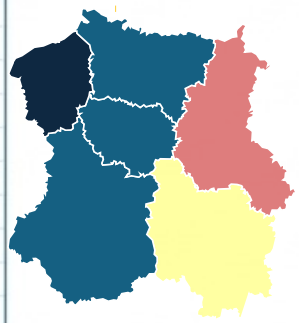
District	RE t=1	RE t=2	RE t=3
1	S_{11}	S_{12}	S_{13}
2	S_{21}	S_{22}	S_{23}
3	S_{31}	S_{32}	S_{33}
4	S_{41}	S_{42}	S_{43}
5	S_{51}	S_{52}	S_{53}
6	S_{61}	S_{62}	S_{63}

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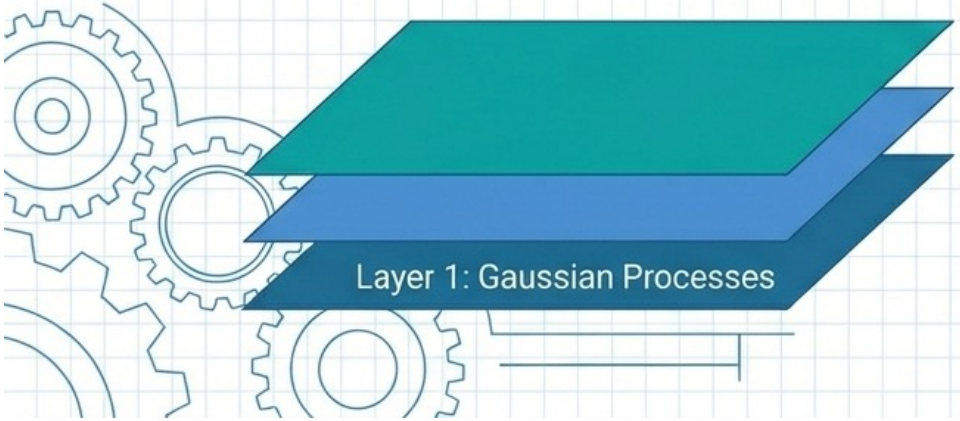


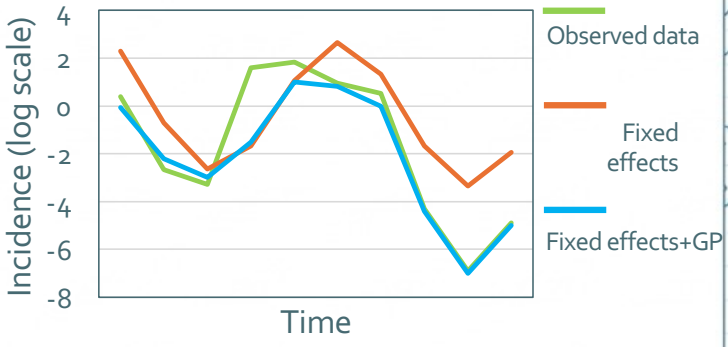


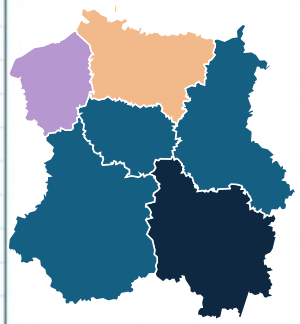
District	RE t=1	RE t=2	RE t=3
1	S_{11} ↑	S_{12} ↑	S_{13} ↓
2	S_{21}	S_{22}	S_{23}
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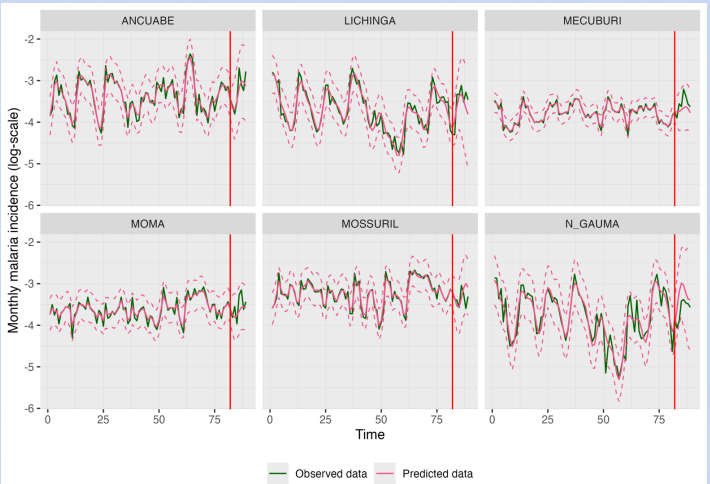
District	RE t=1	RE t=2	RE t=3
1	S_{11}	S_{12}	S_{13}
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4	S_{41}	S_{42}	S_{43}
5	S_{51}	S_{52}	S_{53}
6	S_{61} ↑	S_{62} ↓	S_{63} ↓

Evidence 1: Competitive predictive performance

Knorr-Held!

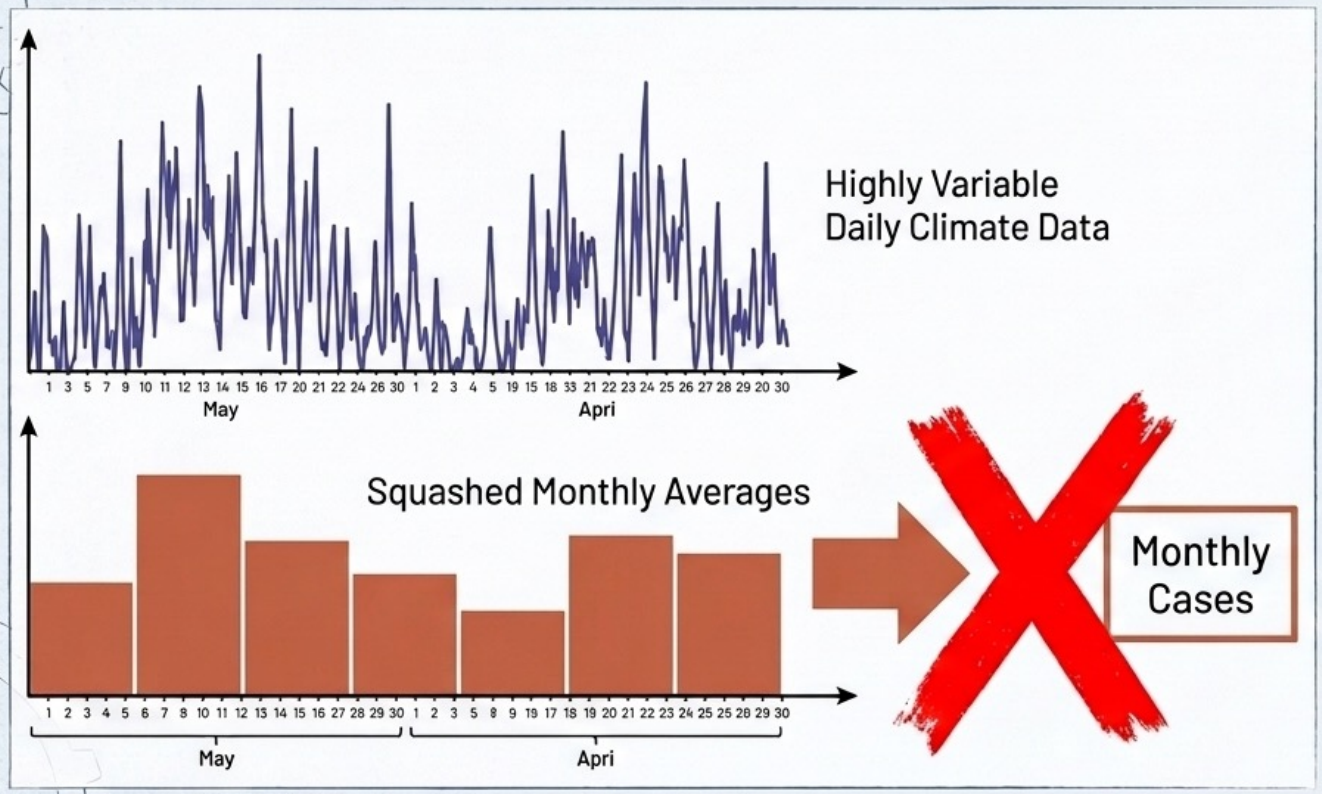
Our proposal!

Model	RMSE	WAIC
M1	2.77 – 3.86	4798
M2	2.59 – 3.79	4690
M3	2.53 – 3.89	4584
M4	1.57 – 3.24	NA
M5	1.88 – 2.98	4506



By introducing spatial dependence in the hyperparameters, the model impose spatial correlation at the parameter level while providing a more flexible representation of temporal dynamics

Problem 2: Temporal misalignment

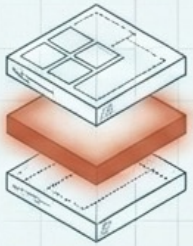


The Limitation

Averaging highly variable daily environmental data into monthly value may destroy crucial short-term environmental signals.

The Question

How can monthly observed disease counts be accurately linked with high-resolution daily climate variables?



Solution 2: Temporal Disaggregation via Latent Incidence

Input

Daily Climate Variables (Temp, Rainfall, Humidity)

Latent Engine

Model predicts unobserved Daily Malaria Incidence (Z_{itd})

Aggregation

$\sum$

The daily incidences are mathematically summed across the month's days

Output

Linked to Observed Monthly Cases (Y_{it})

Instead of aggregating the environment to match the disease, we model the disease at the daily resolution of the environment, and aggregate the predictions to match the observations.

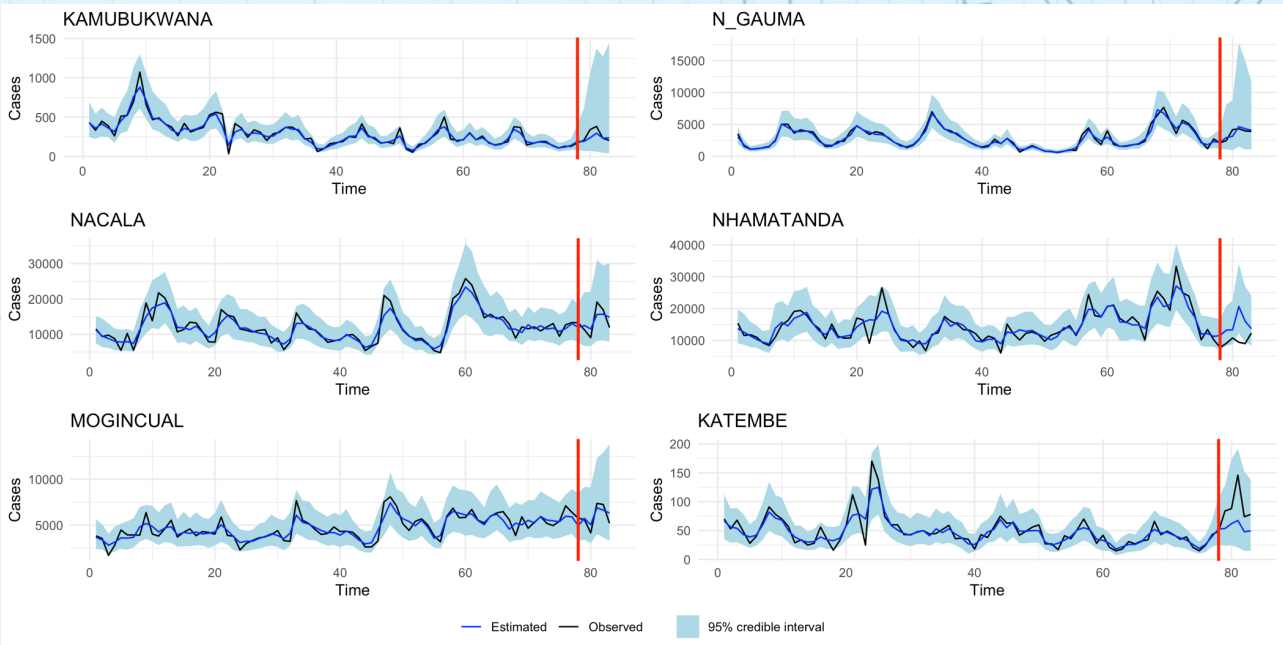
Evidence 2: Enhanced Predictive fit

Baseline: No temporal disaggregation

WAIC: 198422
Training RMSE: 1039
Test RMSE: 3602

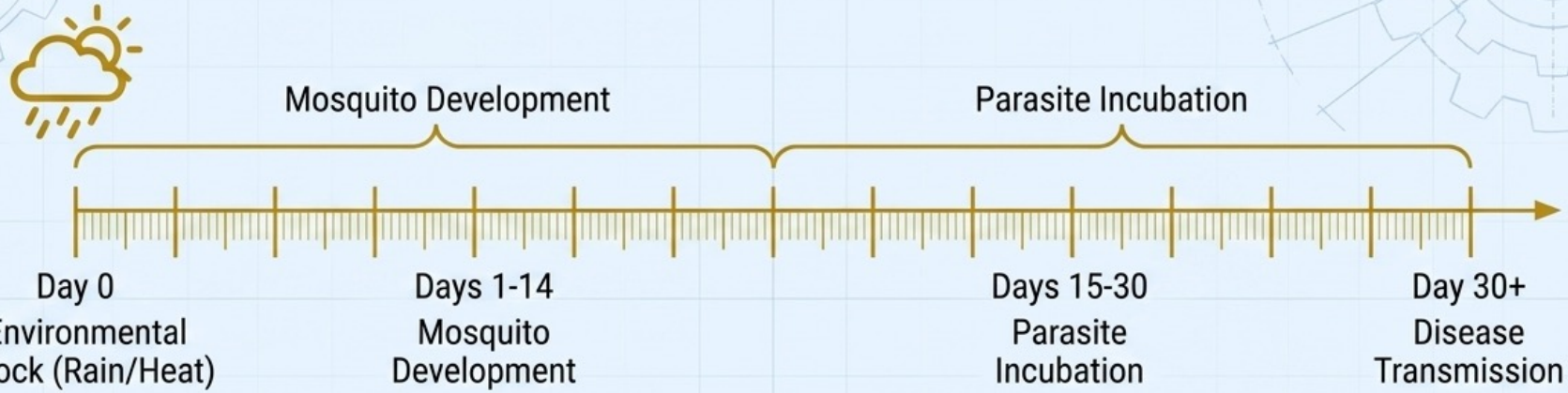
Upgraded: Temporal disaggregation

WAIC: 197379
Training RMSE: 988
Test RMSE: 3284



Temporal disaggregation makes better use of available daily climate information, producing good prediction performance.

Problem 3: Modeling Delayed Biological Realities



The Limitation

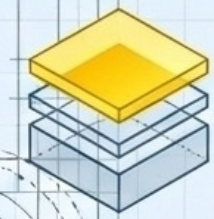
An environmental event today doesn't cause a malaria case today. The effect is delayed and non-linear.

The Decomposition Need

Furthermore, we must separate a district's baseline environment (e.g., average elevation) from short-term weather anomalies.

The Question

How can delayed climatic effects be estimated accurately while accounting for ecological bias?



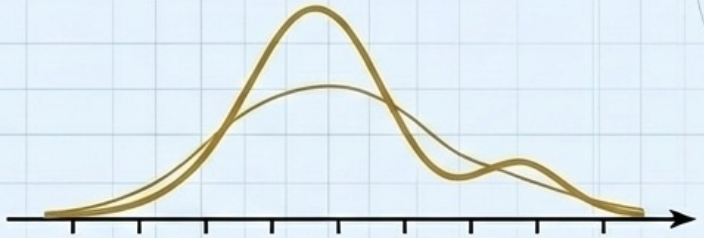
Solution 3: Within-Between Decomposition framework & Bayesian DLNM

The Between Components



These act as spatial baseline drivers (e.g., higher average NDVI = higher baseline risk).

The Within Components

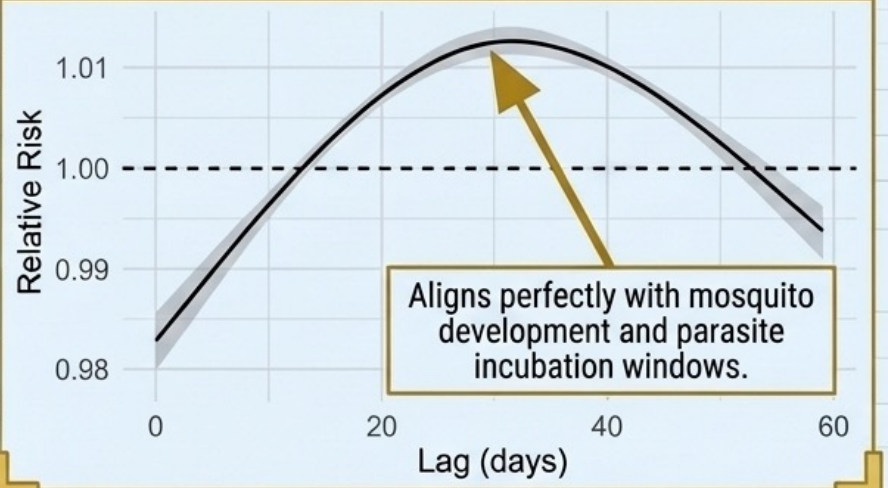


The DLNM isolates short-term anomalies (e.g., hotter than usual for this district) and applies a lag matrix to find delayed effects.

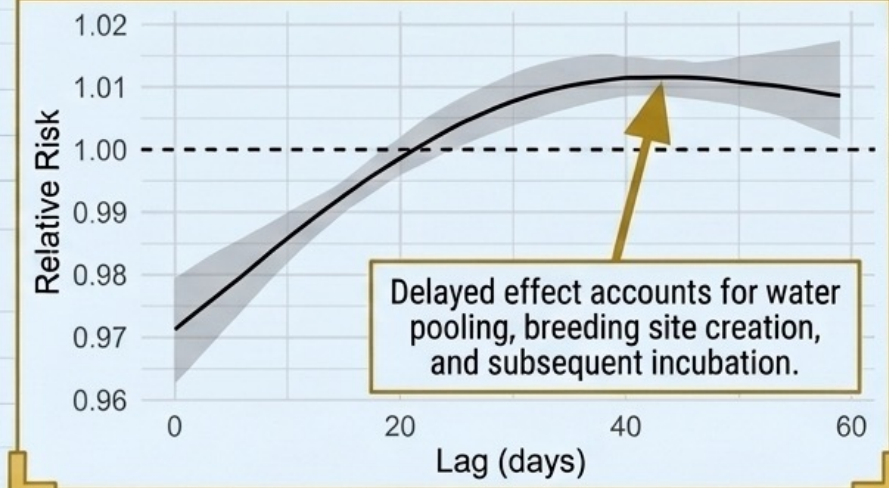
We focus on biological interpretation, not just basis construction. The model isolates local anomalies and tracks their delayed ripple effects through the transmission cycle.

Evidence 3A: Epidemiologically Plausible Lag-Response

Temperature Lag-Response

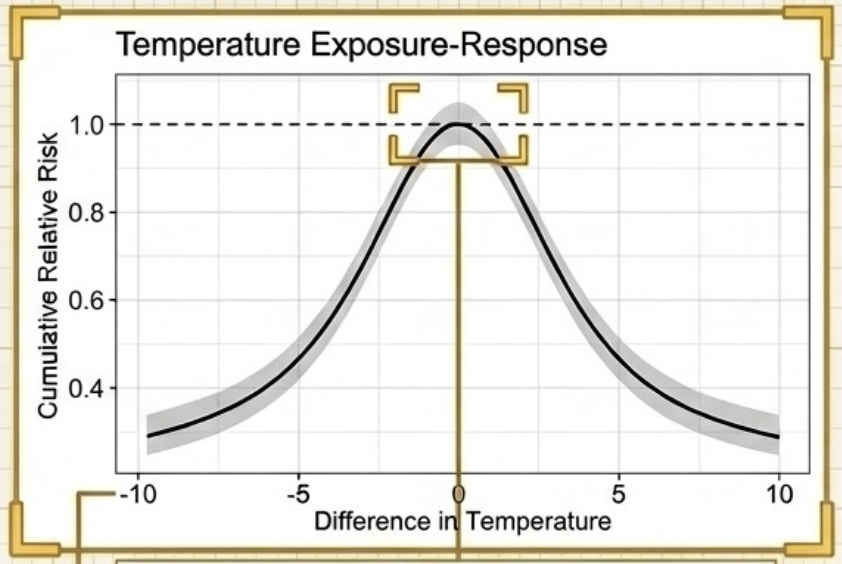


Precipitation Lag-Response

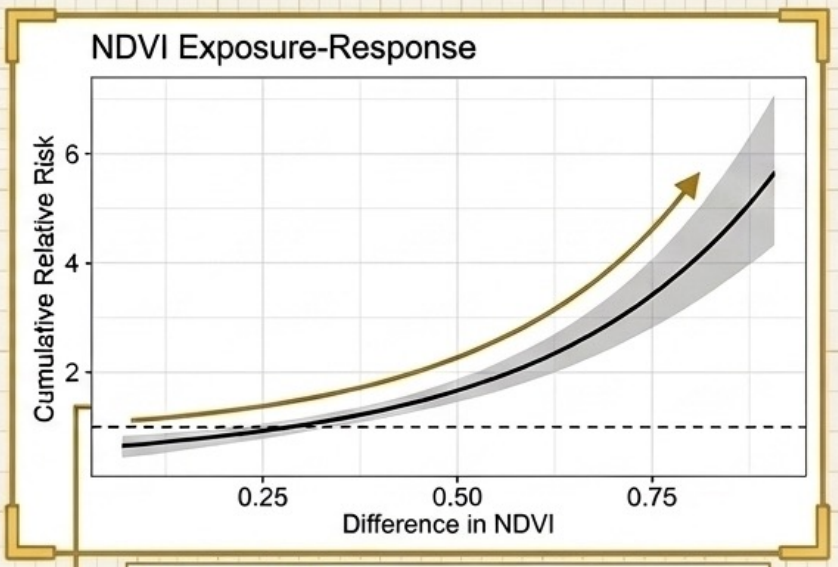


The statistical framework discovers lag structure that match known biological transmission mechanics.

Evidence 3B: Credible Exposure-Response Thresholds



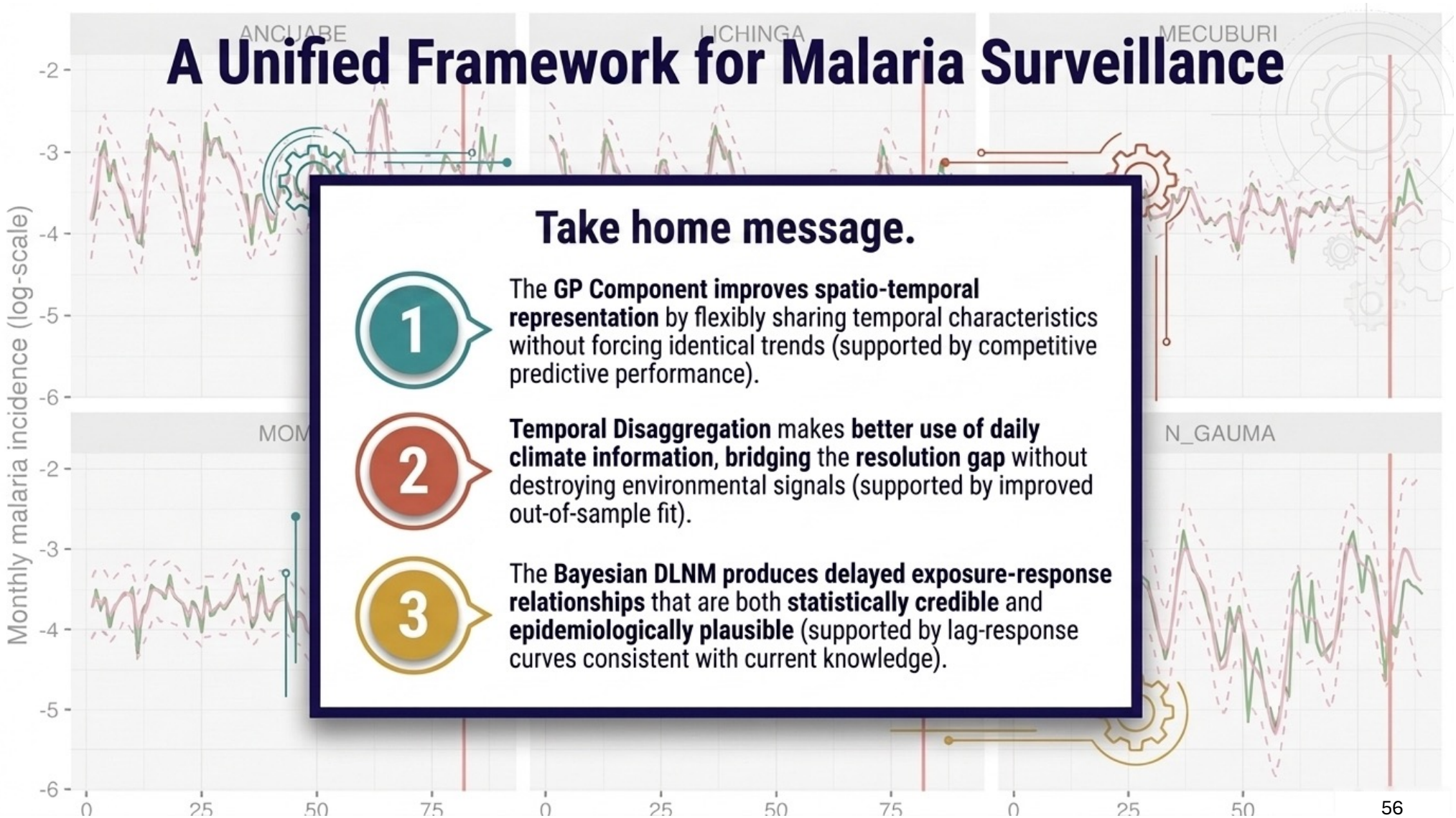
Optimal Transmission Window: Risk drops sharply when temperature deviates too far above or below the local norm.



Exponential Risk: Greener-than-average conditions (high NDVI anomaly) heavily favor elevated mosquito density.

The Bayesian DLNM captures the complex, non-linear realities of disease ecology, ensuring predictions are grounded in plausible biology.

A Unified Framework for Malaria Surveillance



Take home message.

1. The **GP Component improves spatio-temporal representation** by flexibly sharing temporal characteristics without forcing identical trends (supported by competitive predictive performance).

2. **Temporal Disaggregation** makes **better use of daily climate information**, **bridging** the **resolution gap** without destroying environmental signals (supported by improved out-of-sample fit).

3. The **Bayesian DLNM produces delayed exposure-response relationships** that are both **statistically credible** and **epidemiologically plausible** (supported by lag-response curves consistent with current knowledge).